PaDiM Model Optimization

From Research Code to Production-Ready Inference

77x Performance Improvement

March 2026

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What is PaDiM & Why Does It Matter?

How It Works:

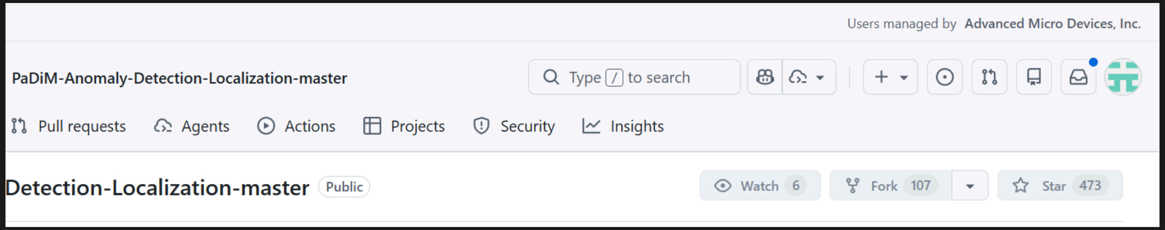
1. **Learn "Normal"**: Extract features from +ve examples using CNN backbone
2. **Build Statistical Model**: Fit Gaussian distribution per patch location
3. **Detect Anomalies**: Measure deviation using Mahalanobis distance

Why PaDiM is Important:

- **Zero-shot anomaly detection** - no -ve samples needed
- **Used in** Manufacturing QC, PCB inspection, Surface anomaly detection
- State-of-the-art - 96%+ AUROC on MVTec AD benchmark

The Challenge:

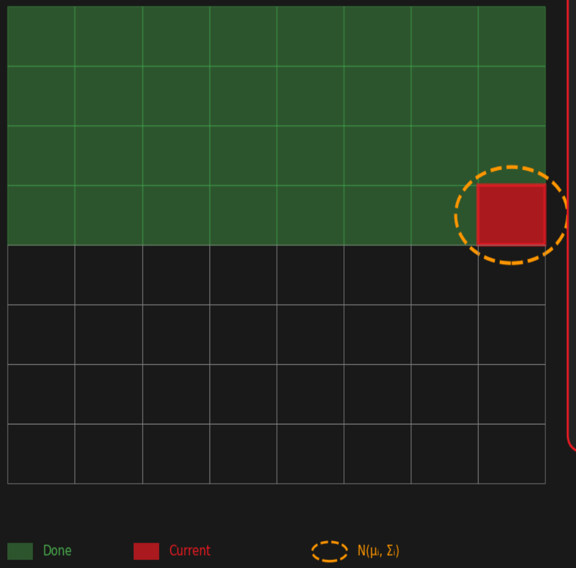
- Implementation from popular open-source repo
- **Not production-ready** out of the box
- **Not designed for edge compute**
- Sequential processing, Python overhead



PaDiM Pipeline: Gaussian Scanning Process

BEFORE: Sequential Loop

Position $i = 32$ of 3,136



LOOP ALGORITHM

For each position $i = 1, 2, \ldots, 3136$:

1. Extract feature vector x_i
2. Get learned mean μ_i
3. Get covariance matrix Σ_i
4. **Compute inverse $\Sigma_i^{-1} = \mathrm{inv}(\Sigma_i)$** ← $O(n^3)$
5. Distance $d_i = \sqrt{[(x_i-\mu_i)^T \Sigma_i^{-1} (x_i-\mu_i)]}$

End loop

Inversions computed: **32** of 64

Total operations: **32,000,000**

621 ms per image | 1.6 FPS | TOO SLOW FOR REAL-TIME

Each patch compared against learned Gaussian distribution

Pipeline Steps:

1. Feature Extraction (ResNet18)
2. Embedding Concat (448D)
3. Dimension Reduction (→100D)
4. Gaussian Scan - Compare each patch to learned distribution
5. Anomaly Map - Mahalanobis distance per spatial location

The Bottleneck:

- 3,136 spatial positions
- Each needs distance computation
- Python loop with SciPy calls

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Key Optimization Insight

Inverse Covariance Can Be Pre-computed

Original Approach (Per-Inference):

```
For each of 3,136 positions:
    1. Get covariance matrix Σ
    2. Compute inverse Σ⁻¹  ← SLOW!
    3. Calculate Mahalanobis distance
```

Problem: Computing Σ^{-1} is $O(n^3) = O(100^3)$

Solution:

- Σ^{-1} is **constant** after training
- Compute once, save to disk (~31 MB)
- Load and reuse during inference

Impact: Eliminates 3,136 matrix inversions at runtime

Optimized Pipeline:

With `inv_cov` and `mean` pre-computed and loaded:

```
# Pre-loaded from training
mean = load("mean.pkl")          # [100, 56, 56]
inv_cov = load("inv_cov.pkl")    # [100, 100, 56, 56]

# Inference - no matrix inverse needed!
diff = embedding - mean
dist = mahalanobis(diff, inv_cov)
```

Result: The expensive operations are moved to training time. Rest of the loop can be **parallelized** and **converted to ONNX**.

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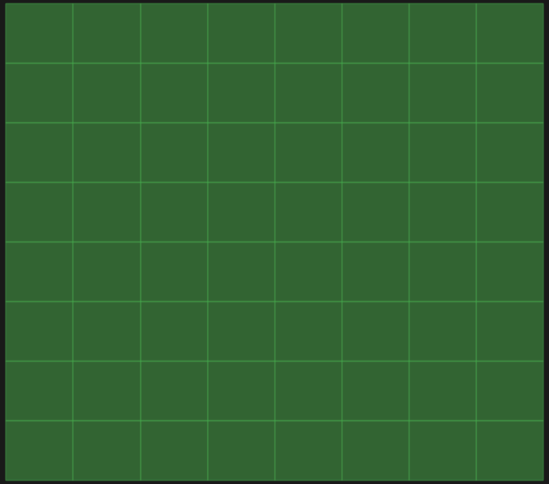
Vectorized Mahalanobis Distance

AFTER: Vectorized Parallel

ALL 3,136 positions at once



Parallel → Single operation

VECTORIZED ALGORITHM

Pre-compute Σ^{-1} during training (once)

1. Compute all differences: $D = X - \mu$
2. Multiply all at once: $T = \Sigma^{-1} \cdot D$
3. Dot product all at once: $d^2 = D \cdot T$
4. Square root: $d = \sqrt{(d^2)}$

No loop • No runtime inverse

Inversions at runtime: **0** (pre-computed)

Tensor operations: **2** parallel ops

BEFORE: **621 ms** → AFTER: **8 ms** **77× FASTER**

Implementation (Einstein Summation):

```
left = np.einsum('ci,cdij->di', diff, inv_cov)
dist = np.sqrt(np.einsum('di,di->i', diff, left))
```

- **Mahalanobis:** $d(x) = \sqrt{(x-\mu)^T\Sigma^{-1}(x-\mu)}$
- **Vectorized:** $\mathrm{dist}[i] = \sqrt{\sum_{c,d} \mathrm{diff}[c,i] \cdot \Sigma^{-1}[c,d,i] \cdot \mathrm{diff}[d,i]}$
- **Key Insight:**
  - No Python loops
  - BLAS/LAPACK optimized, CPU SIMD (AVX2)
  - **ONNX convertible**
  - All 3,136 distances in ONE op

Performance Results

Speed Gains from Baseline to Full ONNX GPU

Configuration	Time (ms)	FPS	vs Baseline
Baseline PyTorch (scipy loop)	621.51	1.6	1.0x
Pre-computed Σ^{-1}	~80	~12	~8x
+ Vectorized einsum (CPU)	36.22	27.6	17.2x
+ ONNX Runtime (CPU)	15.86	63.1	39.2x
+ GPU (MIGraphX)	**8.01**	**124.8**	**77.6x**

Key Achievement: From 621 ms to 8 ms — **77x speedup** with zero accuracy loss

Future Optimization Opportunities

What Could Be Done Next

NPU Acceleration:

- Running CNN backbone on **NPU** could improve performance
- AMD XDNA / Intel NPU support via ONNX Runtime
- Offload ResNet18 feature extraction
- Free up GPU for other tasks

Mahalanobis Distance Optimization:

- Explore more optimal versions of `Mahalanobis Distance` compatible for **NPU/GPU**
- Custom kernels for einsum operations
- Fused operations to reduce memory bandwidth

Other Opportunities:

Technique	Potential	Complexity
NPU backbone	1.5-2x	Medium
INT8 quantization	1.5-2x	Medium
Batch inference	Nx for N images	Low
Custom HIP kernel	1.2x	High

Current Status:

- 125 FPS already exceeds real-time requirements
- Further optimization for edge deployment scenarios

Summary

77x Faster | 8ms Inference | 125 FPS

Metric	Baseline	Optimized
Time	621 ms	**8 ms**
FPS	1.6	**125**
Speedup	-	**77x**
Accuracy	90.5%	**90.5%**

Key Takeaways:

- Pre-compute inverse covariance at training time
- Vectorize with einsum for parallel computation
- ONNX + GPU acceleration for production deployment

Questions?